MLS Salary, Roster Construction, and Team Performance

Dominic Hoar-Weiler (dhoarwei) Brendan Flynn (bflynn)
Nick Kourakis (nkouraki)

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1 Abstract

Major League Soccer runs under strict salary and roster rules, but teams still vary a lot in how well they perform. This project looks at how teams spend their money and whether different roster construction strategies actually lead to better results. Using publicly available MLS salary and performance data, we study how total payroll, salary distribution, and positional spending relate to outcomes like expected goal differential and points per game. We compare several statistical models that each capture a different view of roster construction. Across these approaches, we find that spending more does not guarantee success. Instead, teams that get more performance out of their salary tend to perform better. This suggests that efficiency, not just spending, is the main driver of success in MLS.

2 Introduction

Major League Soccer is designed to promote parity. Teams operate under a salary cap and a set of roster rules that limit how much they can spend and how they can build their teams. Even with these constraints, there is still a large gap in performance across teams. Some teams consistently compete at the top of the table, while others struggle. This raises a simple but important question about how teams are using their limited resources.

2.1 Problem Statement

MLS teams all face similar financial constraints, but they make very different decisions about how to allocate salary across players and positions. There is no clear empirical answer for which of these strategies actually leads to better performance. Some teams concentrate spending on a few high-paid players, while others spread salary more evenly across the roster.

2.2 Why This Matters

In a capped league, every dollar matters. Poor allocation decisions can limit a team's ability to compete, while efficient spending can create an advantage even without a higher budget. From a sports analytics perspective, this is a broader question about how to allocate limited resources in a way that maximizes output. Understanding this relationship can help explain why some teams outperform others despite similar constraints.

2.3 Research Question

How does salary allocation and roster construction relate to team performance in MLS?

More specifically, we study whether team payroll, salary concentration, and positional spending patterns are associated with outcomes such as expected goal differential, goals added, or points per game.

We use publicly available MLS salary and performance data to evaluate multiple statistical models. Each individual model addresses the same project question from a different angle, allowing us to compare which modeling framework is most informative for understanding roster construction strategy.

2.4 Preview of Results

Our results show that higher payroll alone does not explain team success. Instead, the key factor is how efficiently teams convert salary into player performance. Teams that get more value out of their spending tend to perform better than teams that simply spend more.

3 Data

We combine MLS salary information with team- and player-level performance data.

- Salary data: TODO
- Team performance data: TODO
- Player performance data: TODO
- Seasons included: TODO

3.1 Unit of analysis

We use two related units of analysis depending on the model:

- Team-season for team-level models
- Player-season for player-level models

3.2 Variables

3.2.1 Outcome Variables

Potential response variables include:

- Team expected goal differential (xGD)
- Team goals added differential (g+ diff)
- Points per game
- Player goals added (g+)
- Minutes-adjusted player contribution

3.2.2 Predictors

Potential explanatory variables include:

- Total payroll
- Log payroll

- Share of payroll spent on top 3 players
- Salary concentration index (Gini / HHI)
- Positional spending shares
- Average age / experience (if available)
- Designated player indicators (if available)

3.3 Data Cleaning Overview

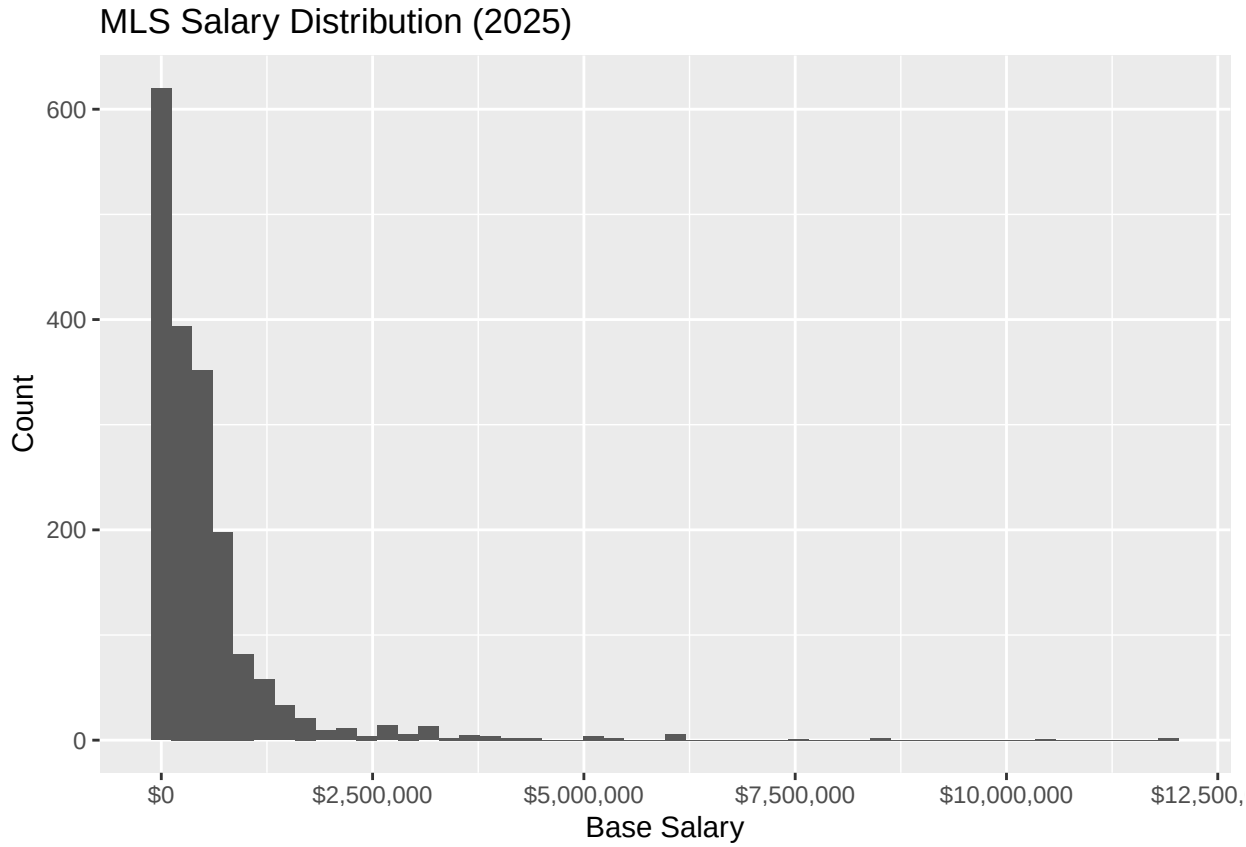
We construct a team-level dataset by aggregating player-level salary and performance data. Key steps include:

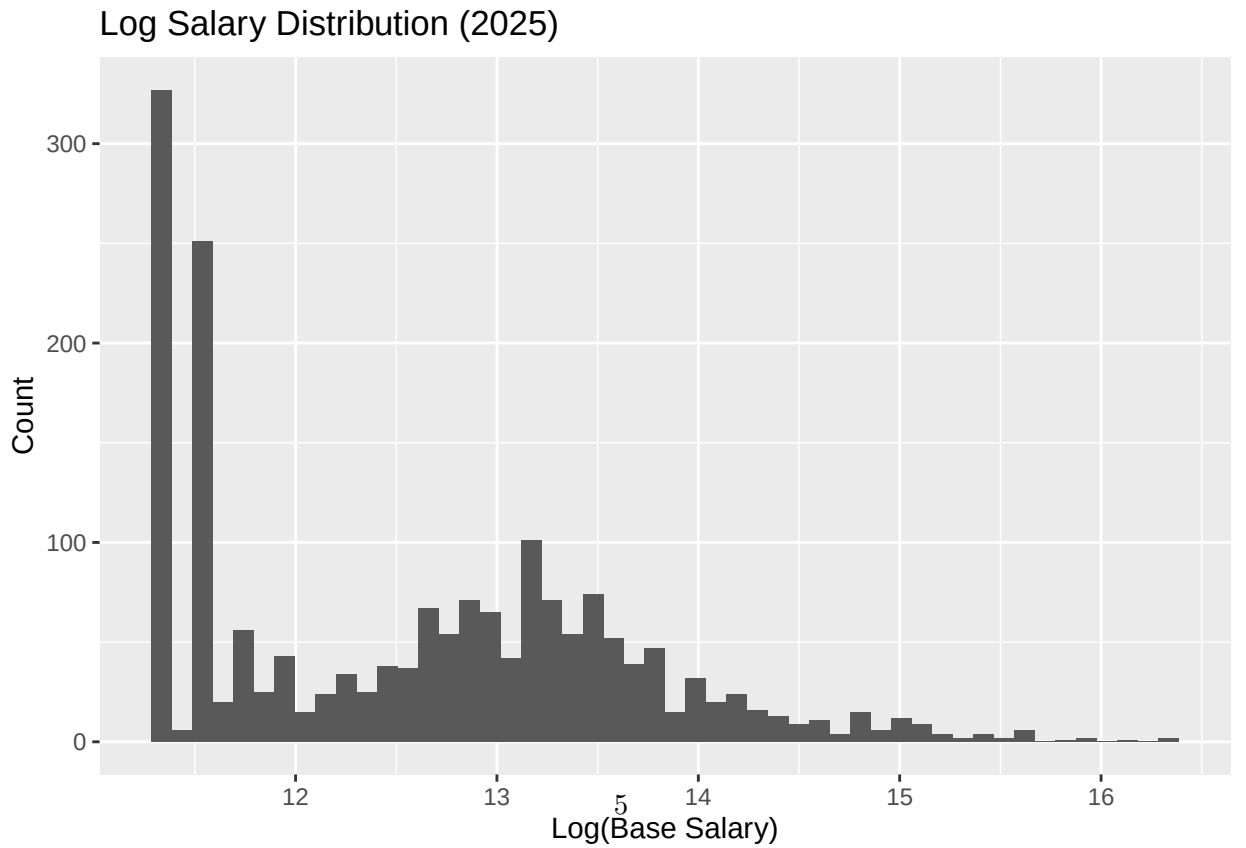
- Aggregating goals added into team-level performance
- Computing salary concentration (top 3 share)
- Creating proxies for designated players using top salary percentiles
- Constructing age-based U22 indicators
- Engineering features capturing roster balance (variance, low-salary share)

These transformations allow us to map individual salary structures to team-level outcomes.

4 Exploratory Data Analysis (EDA)

4.1 Salary distribution

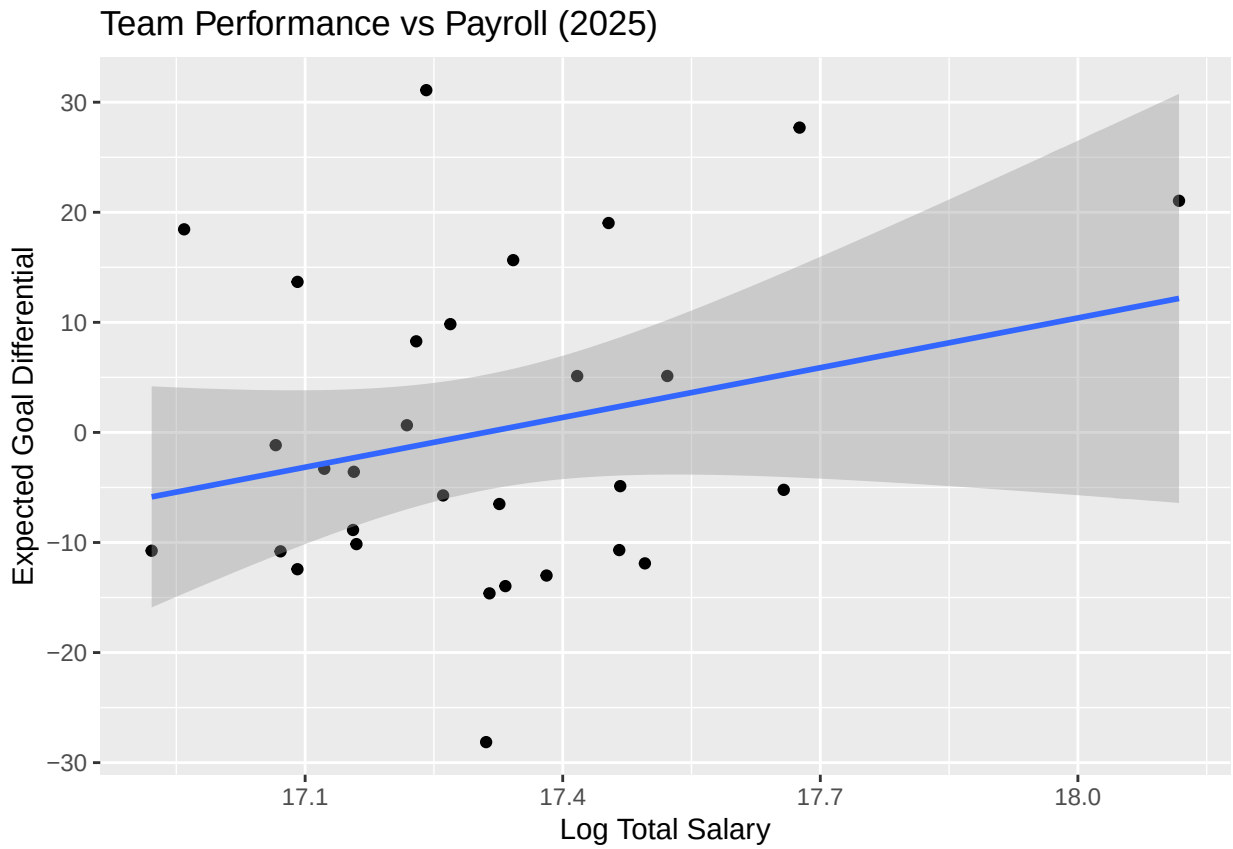




4.2 Takeaway

Salaries are highly right-skewed, meaning a small number of players earn disproportionately high salaries and suggesting a star-driven league structure.

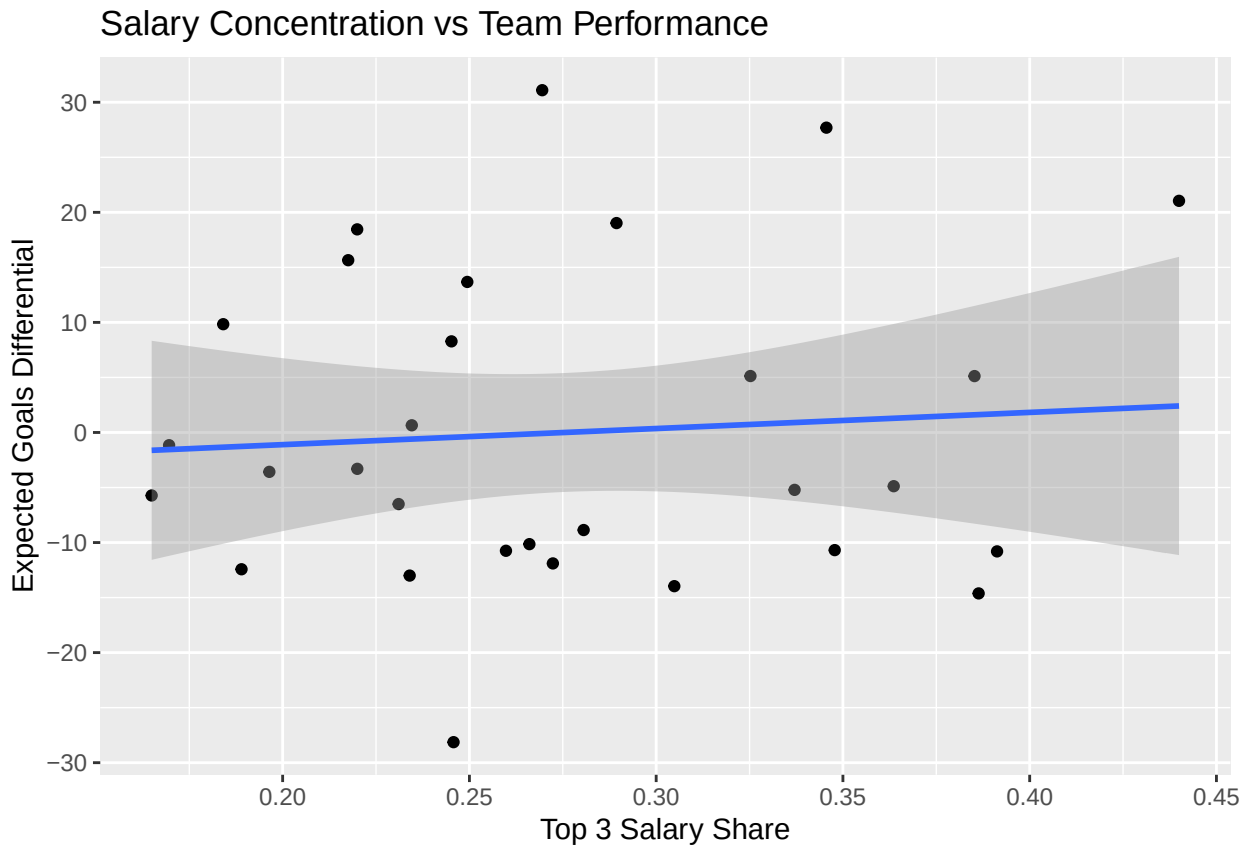
4.3 Team Payroll vs Team Performance



4.3.1 Takeaway

We see a weak/moderate positive relationship (likely). Payroll alone does not fully explain performance, which motivates out decision to add roster structure variables.

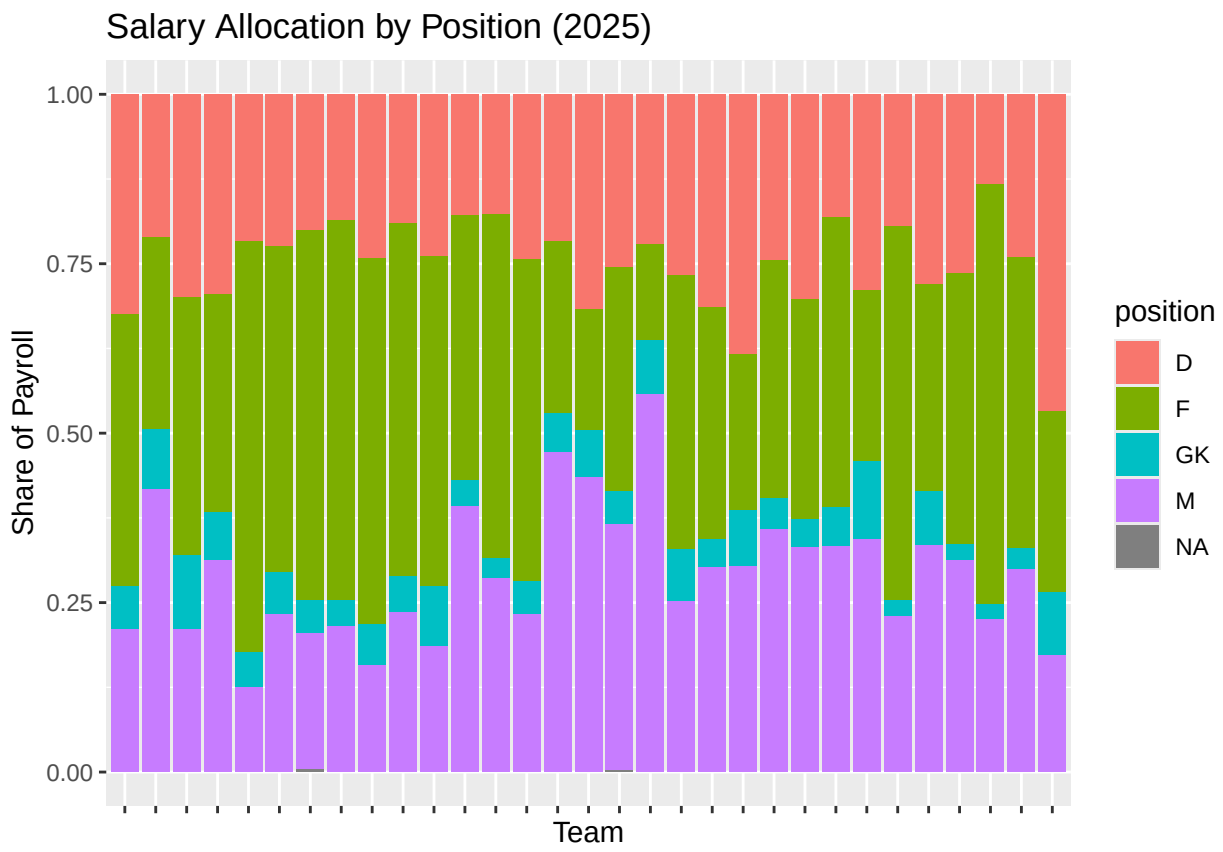
4.4 Salary concentration vs team performance

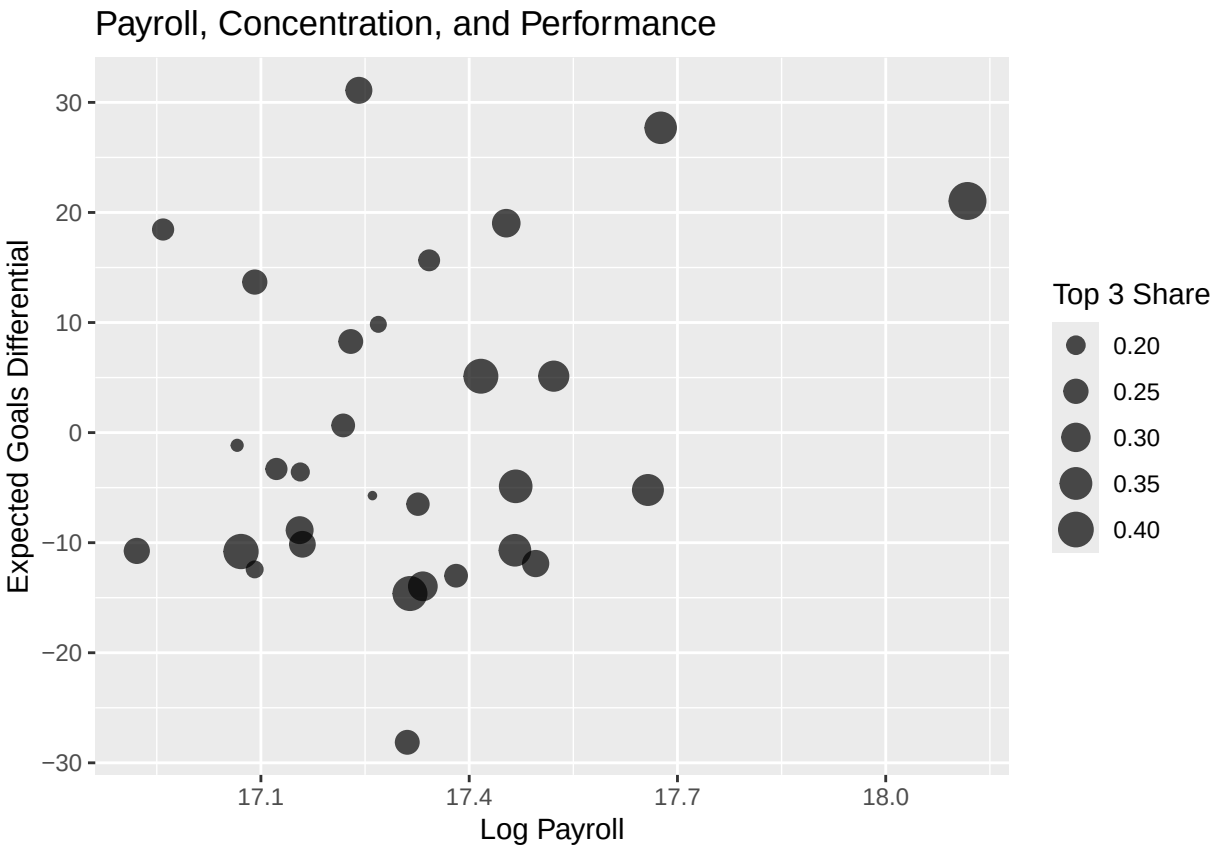


4.4.1 Takeaway

The regression line is essentially flat. Confidence band is wide. Points are highly scattered. No visible monotonic trend.

4.5 Positional Allocation Patterns




Payroll, Concentration, and Performance
Expected Goals Differential
30
20
10
0
−10
−20
−30
17.1
17.4
17.7
18.0
Log Payroll
Top 3 Share
0.20
0.25
0.30
0.35
0.40

5 Methods

Across all individual analyses, the goal is to evaluate how roster construction and salary allocation relate to team success. Each group member fits a distinct model, motivated by a different statistical perspective.

5.1 Individual Analysis 1: Team-season multilevel model (Dominic Hoar-Weiler)

5.1.1 Question

Does team performance increase with total payroll, and does salary concentration help or hurt after controlling for total spending?

5.1.2 Model specification

The final model is a team-season multilevel linear regression with a random intercept for team. The outcome is expected goal differential, and the main predictors are predicted roster value, total payroll, average age, salary concentration, U22 roster investment, and positional salary allocation. The random team intercept allows each MLS club to have its own baseline performance level, accounting for persistent team-level differences such as front-office quality, coaching, market effects, or organizational structure that are not directly observed in the dataset.

5.1.2.1 Baseline Model

$$xGD_{it} = \beta_0 + \beta_1 \log(\text{Salary}_{it}) + \gamma_t + \varepsilon_{it}$$

This model tests whether total payroll alone explains variation in team performance, controlling for season-level effects.

5.1.2.2 Player Level GAM

$$GAAR_{ijt} = f(\log(\text{Salary}_{ijt} + 1)) + \delta_{\text{pos}(j)} + f_{\text{pos}(j)}(\log(\text{Salary}_{ijt} + 1)) + \varepsilon_{ijt}$$

We estimate a nonlinear relationship between salary and player value (goals added above replacement), allowing this relationship to vary by position.

5.1.2.3 Team Level Construction

$$\widehat{V}_{it} = \sum_{j \in i,t} \widehat{GAAR}_{ijt} \cdot \frac{\text{Minutes}_{ijt}}{90}$$

We aggregate predicted player contributions to construct a team-level measure of roster value.

5.1.2.4 Team Multilevel Model

$$xGD_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2)$$

$$
\begin{aligned}
\mu_{it} = \beta_0 &+ \beta_1 \widehat{V}_{it} + \beta_2 \,\text{Salary}_{it} + \beta_3 \,\text{Age}_{it} \\
&+ \beta_4 \,\text{Top3Share}_{it} + \beta_5 \,\text{U22ProxyCount}_{it} \\
&+ \beta_6 \,\text{ForwardShare}_{it} + \beta_7 \,\text{MidfieldShare}_{it} \\
&+ u_i
\end{aligned}
$$

$$
u_i \sim \mathcal{N}(0, \sigma^2_{\text{team}})
$$

We include team-level random effects to account for unobserved differences across teams that persist over time. Predicted roster value, salary, and average age are standardized before fitting the multilevel model, so their coefficients represent changes in expected goal differential for a one-standard-deviation increase in each predictor.

5.1.3 Assumptions

Across models, we assume:

- Linearity in parameters for the regression components
- Residual variation is normally distributed around the fitted mean
- Errors satisfy $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$
- Conditional on the included covariates and team random effects, residual errors are independent across team-seasons.
- Constant variance of errors
- The included roster construction variables capture the main observed differences relevant to the project question, while remaining unobserved team differences are partially absorbed by the team random intercept

For the multilevel model, we additionally assume:

- $xGD_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2)$
- Random effects follow $u_i \sim \mathcal{N}(0, \tau^2)$ and are independent of ε_{it}

5.1.4 Why this model is appropriate

This modeling approach directly matches the project question by separating two effects: how much teams spend and how efficiently they convert that spending into performance. The baseline model provides a simple benchmark, while the player-level GAM and final multilevel model incorporate roster construction features and player-level estimates. The GAM allows for flexibility in the salary-performance relationship at the player level, which avoids imposing a strict linear assumption. The multilevel model accounts for persistent team-level differences that are not captured by observed variables.

5.1.5 Model Evaluation Plan

I evaluate the model progression using in-sample RMSE and MAE as descriptive measures of predictive error. I compare simpler payroll-only and fixed-effect team models against the final multilevel model to assess whether adding player-value and roster-construction variables improves fit. I also use residual diagnostics to check for obvious violations of model assumptions, including nonlinearity and heteroskedasticity.

5.1.6 Uncertainty Plan

I quantify uncertainty using 95% confidence intervals for the estimated coefficients. These intervals show how much uncertainty remains around each estimated effect and help assess whether the data provide clear evidence of positive, negative, or near-zero relationships. In the multilevel model, uncertainty in team-level heterogeneity is summarized through the estimated random-effect variance component.

5.1.7 Other Modeling Notes

In preliminary team-level models, pooling all seasons without controlling for year effects made salary variables difficult to interpret. After introducing season fixed effects in the baseline model, total payroll became more informative. This suggests that league-wide changes, such as salary cap growth and evolving roster rules, can confound the relationship between spending and performance if not properly controlled for.

5.2 Individual Analysis 2: Roster Architecture (Nick Kourakis)

5.2.1 Question

Does team performance depend more on the salary disparity between highest and lowest paid starters, or on the the depth of it's TAM-eligible[1] roster?

5.2.2 Model specification

5.2.2.1 Baseline Model

$$xGD_{it} = \beta_0 + \beta_1 Gini_{it} + \varepsilon_{it}$$

5.2.2.2 Dispersion Model

$$xGD_{it} = \beta_0 + \beta_1 \text{Gini}_{it} + \beta_2 \text{MidClassDepth}_{it} + \beta_3 \log(\text{TotalSalary}_{it}) + \varepsilon_{it}$$

Evaluates the importance of TAM players, controlling for total team spending.

5.2.2.3 Multi Season Roster Model

$$xGD_{it} = \beta_1 \text{Gini}_{it} + \beta_2 \text{MidClassDepth}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Within a specific team over several years, does performance improve when that specific team moves toward a more balanced roster?

5.2.3 Assumptions

- `Linearity`
- `No Multicollinearity`
- `Independence`

5.2.4 Why this model is appropriate

Due to the MLS salary cap, every dollar spent on a "Star" is a dollar that could've been used somewhere else on the roster. The previous analysis evaluates spending efficiency, this model tests the roster structure. By using the Gini coefficient and TAM-depth, we can determine if success comes from a roster that is top-heavy (high inequality) or from a deep roster (high middle-class density).

5.2.5 Model evaluation plan

- `Adjusted $R^2$: To measure how much additional variance in xGD is explained by roster distribution compared to a baseline model.`
- `Akaike Information Criterion (AIC): To determine if adding "Middle Class" complexity improves model fit without over-fitting.`
- `Coefficient Analysis: Evaluating the significance of $\beta_1$ (Gini) to identify whether inequality is correlated with performance.`

[1] TAM (Targeted Allocation Money) Players: The MLS "middle class," consisting of high-level starters earning above the Maximum Salary Budget Charge ($803,125) but below the Designated Player threshold. These players' budget hits are "bought down" using league-allocated funds. See the official documentation here.

5.2.6 Uncertainty plan

- 95% confidence intervals: To verify that the effects of Gini and TAM-depth are statistically distinguishable from zero.
- Clustered Standard Errors: Grouped by team_id to account for correlation in performance across multiple seasons.
- Sensitivity Analysis: Testing the model with and without extreme salary outliers.

5.3 Individual Analysis 3 (Brendan Flynn)

5.3.1 Question

5.3.2 Model specification

5.3.3 Assumptions

5.3.4 Why this model is appropriate

5.3.5 Model evaluation plan

5.3.6 Uncertainty plan

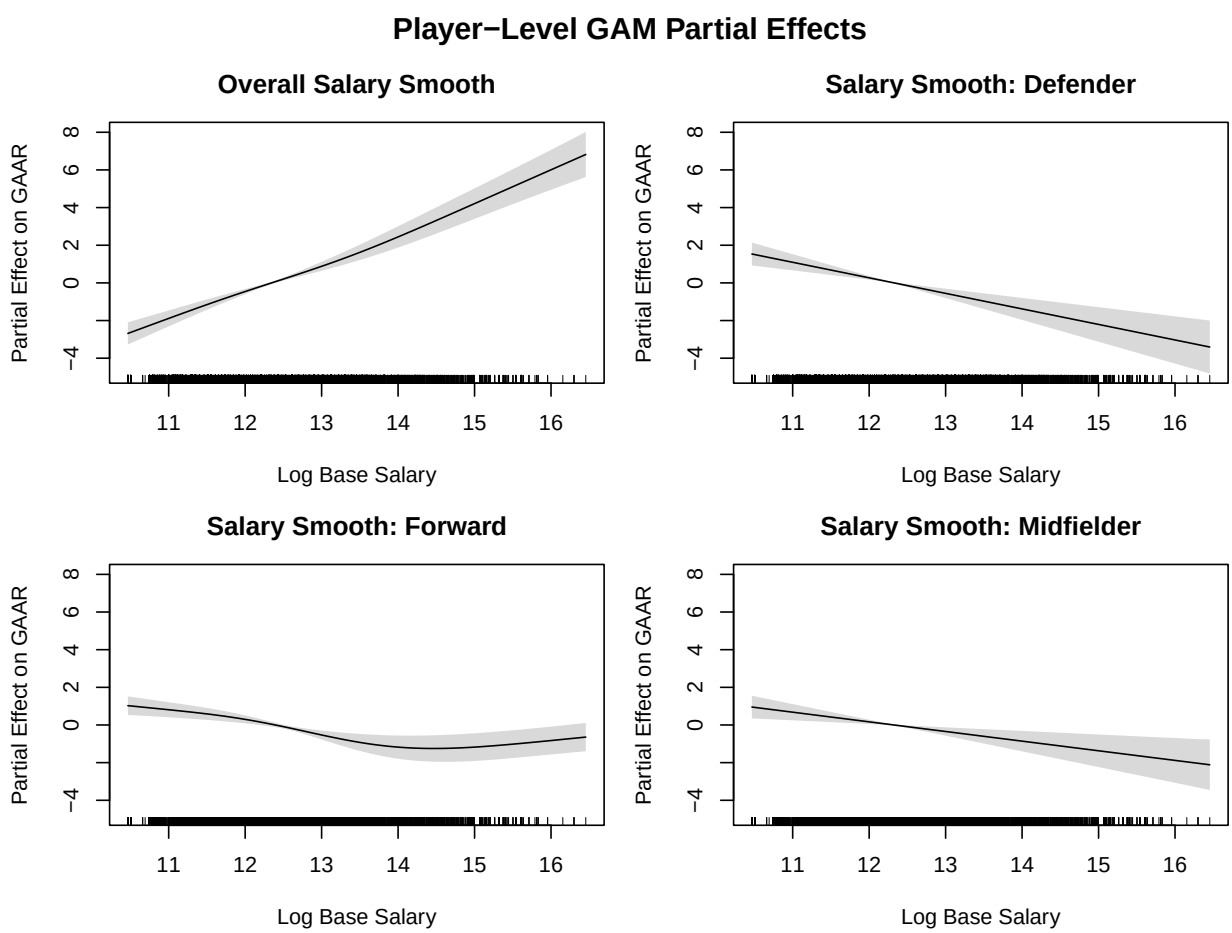
6 Results

6.1 Individual Analysis 1 Results (Dom)

The results below focus on the player-level GAM and the final multilevel model. The GAM is used to construct predicted player value, while the multilevel model is used for the main team-level inference.

Table 1: Final Multilevel Model: Fixed Effects

term	estimate	std.error	p.value	conf.low	conf.high
Intercept	-10.861	6.005	0.071	-22.676	0.954
Predicted roster value	6.519	0.982	0.000	4.587	8.452
Total salary	-6.008	1.161	0.000	-8.293	-3.723
Average age	-1.207	0.707	0.089	-2.598	0.184
Top 3 salary share	13.747	7.208	0.057	-0.437	27.931
U22 proxy count	-0.343	0.567	0.546	-1.459	0.773
Forward salary share	6.083	9.707	0.531	-13.018	25.184
Midfield salary share	7.296	9.217	0.429	-10.841	25.434



6.1.0.1 Player Level GAM

The player-level GAM shows that salary is related to goals added above replacement, but the relationship is not the same across positions. The overall salary smooth is positive, meaning that higher-paid players generally have higher predicted value. The position-specific smooths show how this overall salary effect is adjusted for defenders, forwards, and midfielders.

For defenders and midfielders, the position-specific smooths decline as salary increases, suggesting that the salary-value relationship is weaker for those positions than the overall trend. For forwards, the partial effect decreases until roughly a log base salary of 14, or about $1.2 million, before increasing slightly at higher salaries. This suggests that salary is useful as a signal of player value, but the return to spending depends on positional context.

These results support using predicted player value rather than raw salary in the team-level model. Payroll alone does not tell us whether a team is actually getting production from its spending, so the GAM helps translate player salaries into estimated on-field value before aggregating to the team-season level.

6.1.0.2 Multilevel Team Model

The main result is that predicted player value is strongly positive ($\beta = 6.52, p < 0.001$), while total salary is negative conditional on value ($\beta = -6.01, p < 0.001$). These two predictors are correlated by construction; predicted value is derived from salary, so their coefficients should be interpreted jointly: teams that convert spending into production outperform teams that spend without generating value, not that spending itself is harmful.

Salary concentration has a positive point estimate, but the confidence interval includes zero (95% CI: -0.44, 27.93), so this should be treated as suggestive rather than conclusive.

The primary finding is that efficient roster construction (converting salary into player value) matters more than total payroll. Positional salary shares and U22 investment do not show independent effects once predicted value and team heterogeneity are accounted for.

6.1.1 Model Comparison

Table 2: Model Comparison

Model	RMSE	MAE
Payroll-only benchmark	11.32	8.89
Fixed-effect team model	9.51	7.49
Final multilevel model (in-sample)	9.14	7.26
Final multilevel model (LOOCV)	10.12	8.01

In-sample error decreases across the three models, with the final multilevel model achieving the lowest RMSE and MAE. LOO-CV estimates for the final model (RMSE = 10.12, MAE = 8.01) are modestly higher than in-sample, which is expected given the small sample (about 29 team-seasons). The limited gap between in-sample and LOO-CV suggests the model is not badly overfit, though out-of-sample performance would need to be validated on additional seasons before drawing strong predictive conclusions.

6.1.2 Key Takeaways

The strongest result is that predicted player value is the best predictor of team performance. Once predicted player value is included, salary has a negative conditional effect. This supports an efficiency interpretation: teams perform better when they convert spending into actual production.

There is also some evidence that salary concentration may help, but this result is less strong than the player value and salary findings. The U22 proxy and positional salary shares do not provide clear evidence of independent effects in the final model.

6.1.3 Answer to Research Question

The results suggest that team performance in MLS is related to payroll, but not simply because spending more automatically produces better teams. At the team level, performance is explained more by predicted player value than by raw salary. Conditional on value, higher salary is associated with worse outcomes, suggesting that inefficient spending hurts roster construction.

Salary concentration may help, but this result is less certain than the player value and salary findings. The GAM also suggests that salary does not translate into value in the same way across positions, which supports using predicted player value rather than raw payroll as the main roster-quality measure. However, the final multilevel model does not show clear independent effects for positional salary shares, so the main conclusion is about efficiency rather than simply shifting spending toward a specific position.

Overall, efficient roster construction appears to matter more than total spending alone.

6.2 Individual Analysis 2 Results (Nick)

6.3 Individual Analysis 3: Positional Spending (Flynn)

6.3.0.1 Question Does the allocation of salary across positions affect team performance? In particular, do teams that spend a larger share of payroll on attacking players, midfielders, defenders, or goalkeepers perform differently on average?

6.3.0.2 Model Specification While total payroll measures the overall financial resources a team invests into their roster, teams vary drastically in what positions they invest more heavily into. Some teams may prioritize investing heavily into forwards who are generally most responsible for increasing the expected goals of a team but also garner a higher salary. Other teams may prioritize a more balanced spending strategy, spreading spending out across positions.

6.3.0.3 Baseline Model

$$xGD_{it} = \beta_0 + \beta_1 FShare_{it} + \beta_2 MShare_{it} + \beta_3 DShare_{it} + \beta_4 \log(Payroll_{it}) + \gamma_t + \varepsilon_{it}$$

6.3.0.4 Assumptions

- Linearity
- Independence
- No Multicollinearity
- Homoskedasticity

6.3.0.5 Why this model is appropriate This regression framework directly addresses the central research question by estimating the marginal effect of reallocating payroll across positions. By including total payroll in the model, we separate the overall level of spending and the distribution of spending across positions. Season fixed effects further control for league-wide changes such as rule changes, league expansion, or shifts in playing style over time. Evaluates whether teams that allocate a larger share of payroll to forwards, midfielders, or defenders perform differently on average, while controlling for season-level effects.

6.3.0.6 Model Evaluation Plan Model performance will be evaluated using several metrics:

- Coefficient significance tests to determine whether positional spending shares are associated with team performance
- R-squared and adjusted R-squared to evaluate the proportion of variation in expected goal differential explained by the model
- Residual diagnostics to check for violations of model assumptions

6.3.0.7 Uncertainty Plan

- standard errors and confidence intervals for regression coefficients
- hypothesis testing using t-statistics and p-values
- sensitivity checks with alternative specification

6.3.1 Clustering Approach

6.3.1.1 Model Specification In addition to the regression model, a clustering approach is used to identify common payroll allocation strategies among MLS teams. Each team-season observation is represented by four features: Forward salary share, Midfielder salary share, Defender salary share, and Goalkeeper salary share. A k-means clustering algorithm groups teams into clusters based on these salary allocation profiles. The algorithm minimizes within-cluster variation in payroll allocation patterns.

6.3.1.2 Assumptions

- Teams with similar salary allocation profiles represent similar roster construction strategies.
- Euclidean distance between salary shares appropriately captures similarity between teams.
- The selected number of clusters adequately represents the underlying strategies present in the data.

6.3.1.3 Why this model is appropriate While regression analysis estimates marginal effects of salary allocation, clustering identifies distinct team-building strategies present in the league. This approach allows us to examine whether teams that emphasize different positional spending strategies exhibit different performance outcomes.

6.3.1.4 Model Evaluation Plan Cluster quality will be evaluated using: - cluster sizes and stability - visual inspection of cluster centers - comparison of average team performance across clusters

6.3.1.5 Uncertainty Plan

Uncertainty in the clustering results will be addressed by:

testing alternative numbers of clusters examining the robustness of cluster centers comparing performance differences across clusters

7 Discussion

Our project examined whether MLS teams benefit more from spending heavily overall, concentrating spending in a few stars, or distributing spending more efficiently across the roster.

8 Appendix

8.1 Section A: Code

Describe any major cleaning steps here